



**MIDDLE EAST TECHNICAL UNIVERSITY
NORTHERN CYPRUS CAMPUS**

Computer Engineering Program

CNG 300

SUMMER PRACTICE REPORT

Name of Student:

ID Number:

Name of Company:

Project Title: Linux System Administration and Database Management in DevOps Environments

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Abstract

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1. ABC Teknoloji Bilişim Çözümleri A.Ş.

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2. Introduction

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3. Problem Statement

4. Solutions

This section describes the systems and tasks I worked with during my internship, including their significance and how I approached them in practice. Where company policy applies, the information is presented in a generalized form (without internal paths, configurations, or proprietary code).

4.1 Linux

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4.1.1 SQL

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4.1.2 Bash

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4.2 Installing MySQL

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4.2.1 Install & start

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4.2.2 Secure the fresh instance

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4.2.3 First look inside MySQL

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4.2.4 Why this setup mattered

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4.3 SQL Problem-Solving in Practice

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4.3.1 Servers with consistently slow backups

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4.3.2 Largest tables in each schema

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4.3.3 Schemas that missed a successful backup in 24 hours

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4.3.4 Replica and primary version drift

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4.4 Read replicas

We worked with a primary database that takes all writes and several replicas that stay read-only.....

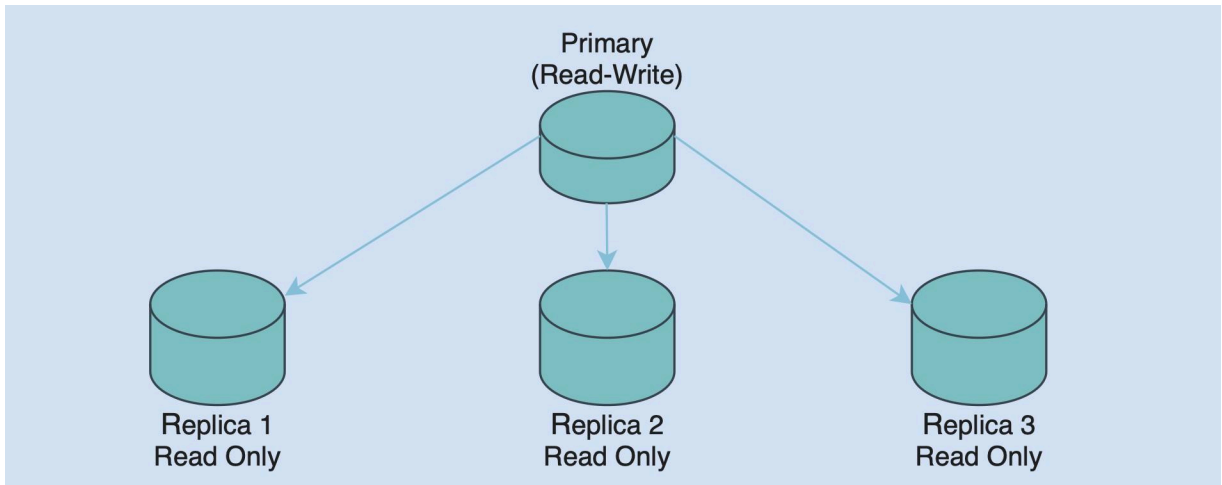


Figure 8: Primary–replica database replication (single-primary).

4.4.1 Why we run with replicas

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4.4.2 What we look at in cluster status

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4.4.3 Illustrative cluster status

.....

4.4.4 What each field told us and why we check it.

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4.4.5 How we validated the setup

.....

4.5 Backup, restore, and automation

.....

4.5.1 Daily logical backup

.....

4.5.2 Restore into a clean schema

.....

4.5.3 Sanity checks right after restore

.....

4.5.4 Schedule the job and rotate, and log housekeeping

.....

4.5.7 MongoDB snapshot and restore

.....

4.5.8 MongoDB quick checks

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Conclusion

This summer practice focused on two things that.....

I installed MySQL locally with Homebrew,.....

The main contribution was turning these pieces into a repeatable workflow:
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End result:

Gains were both technical and judgment-oriented.....

References

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